An Adaptive Multi-Sensor Electronic Nose for Autonomous Fire and Gas Hazard Mitigation with Real-Time Machine Learning Telemetry Logging

Abstract—**Conventional microcontroller-based gas and fire detection systems rely on single-sensor raw analog thresholds that are highly vulnerable to analog-to-digital converter (ADC) noise, cross-sensitivity among hydrocarbon species, and long-term baseline drift caused by ambient temperature and humidity fluctuations. Furthermore, traditional embedded alarm circuits operate as isolated loops that discard intermediate signal states rather than generating structured telemetry for data-driven Machine Learning (ML) classifiers. This paper presents the design, mathematical formulation, hardware implementation, and experimental validation of a multi-sensor adaptive electronic nose and autonomous hazard mitigation system built on the Arduino Uno R3 (ATmega328P) that simultaneously performs local physical intervention and streams pre-conditioned 17-feature CSV telemetry for ML model training and inference. The sensing array integrates three metal-oxide semiconductor (MOS) gas sensors (MQ2, MQ5, and MQ7), an active-low infrared (IR) flame detector, a DHT22 digital temperature and relative humidity sensor, and a hardware push-button interface. Following a 50-sample clean-air auto-calibration phase, the firmware executes a dual-stage Exponential Moving Average (EMA) pipeline: a fast signal EMA filter ($\alpha_{\text{signal}} = 0.20$) suppresses high-frequency ADC noise, while a conditional slow baseline EMA filter ($\alpha = 0.01$) continuously compensates for ambient drift during verified quiescent conditions ($|S_{\text{norm}}| < 0.05$) and freezes during gas excursions. A 4-class sensor fusion engine differentiates Carbon Monoxide (CO), LPG/Propane, Smoke/Fire, and Methane/Natural Gas (NG), validated through a** 2000 **ms temporal confirmation filter to eliminate transient false alarms. Upon hazard confirmation, the microcontroller triggers an active buzzer, updates a state-buffered anti-flicker** 16×2 **I2C LCD, and energizes a 5V DC exhaust fan via a flyback-protected 2N2222 NPN switching stage while streaming 17 synchronized telemetry features at** 1 **Hz over** 9600 **baud UART.**

Index Terms—**Adaptive electronic nose, Arduino Uno R3, baseline drift compensation, dual-stage exponential moving average (EMA), fire and gas leakage mitigation, machine learning telemetry, multi-sensor fusion, temporal debouncing.**

. INTRODUCTION

Accidental leaks of combustible hydrocarbons—such as Liquefied Petroleum Gas (LPG), propane, and methane (natural gas)—alongside toxic carbon monoxide (CO) accumulation and incipient structural fires represent critical safety hazards in residential, laboratory, and industrial environments. If unmitigated during early diffusion stages, flammable gas pockets can rapidly exceed their lower explosive limit (LEL), while low concentrations of odorless CO induce acute hypoxia. Consequently, safety engineering demands embedded monitoring architectures capable of rapid detection, gas species differentiation, autonomous physical ventilation, and structured data logging.

Most low-cost embedded detectors reported in existing literature employ a single metal-oxide semiconductor (MOS) gas sensor compared against a static raw ADC threshold [1, 6]. In practical deployments, such architectures suffer from four fundamental limitations:

1. *High-Frequency ADC and Thermal Noise:* Raw analog readings from SnO_2 sensing layers exhibit stochastic cycle-to-cycle fluctuations that trigger false alarms near fixed comparator boundaries.
2. *Cross-Sensitivity Across Gas Species:* Individual MQ-series sensors respond simultaneously to multiple reducing gases (e.g., MQ2 responds to LPG, propane, hydrogen, and smoke), making single-sensor identification impossible.
3. *Baseline Drift and Aging:* Ambient temperature shifts, relative humidity variations, and long-term SnO_2 grain boundary aging cause clean-air baseline resistance (R_0) to drift over time, degrading fixed-threshold sensitivity [4].
4. *Absence of ML-Ready Telemetry:* Conventional microcontroller alarms output only binary triggers or unstructured text strings, discarding the intermediate filtered, baseline, normalized, and environmental states required to train supervised Machine Learning (ML) models.

To address these gaps, this paper presents an enhanced *Multi-Sensor Adaptive Electronic Nose and Autonomous Hazard Mitigation System* implemented on the Arduino Uno R3 platform. The system couples a heterogeneous 5-sensor array (MQ2, MQ5, MQ7, IR flame sensor, and DHT22 temperature/humidity sensor) with a 4-stage embedded signal processing pipeline: (i) 50-sample startup auto-calibration, (ii) fast EMA signal smoothing

($\alpha_{\text{signal}} = 0.20$), (iii) conditional slow EMA baseline drift tracking ($\alpha = 0.01$), and (iv) 4-class normalized pattern fusion gated by a 2000 ms temporal confirmation window. Alongside local actuation (audible buzzer, state-buffered anti-flicker I2C LCD, and 2N2222 NPN transistor-driven exhaust fan with 1N4007 flyback protection), the firmware serializes a 17-column comma-separated values (CSV) feature vector at 1 Hz for downstream ML ingestion.

. Related Work

Baballe *et al.* [1] demonstrated an automatic gas monitoring circuit using an MQ-5 sensor and microcontroller to activate an audible buzzer when raw readings crossed a preset limit. While inexpensive, single-sensor threshold logic lacked cross-gas discrimination and drift immunity. Islam *et al.* [2] integrated IoT cloud notifications and exhaust actuation into a gas and fire detection node; however, their sensing layer relied on unconditioned raw readings without temporal debouncing or multi-feature ML logging.

Adekitan *et al.* [3] and Apeh *et al.* [5] combined LPG sensing with solenoid shutoff valves, exhaust fans, and SMS alerts for kitchen and building evacuation safety. Both studies confirmed that automated ventilation substantially reduces deflagration risk, yet neither incorporated multi-sensor normalization or gas classification. Ayaz and Yüksel [4] experimentally evaluated MQ-series sensors across open, semi-closed, and closed test chambers, proving that sensor linearity and baseline stability depend strongly on calibration routines and ambient temperature/humidity conditions. Khan [6] similarly noted that household sensor-based detectors require intelligent classification and drift compensation to prevent nuisance alarms from cooking vapors. Table I contrasts these prior systems against our proposed adaptive architecture.

. System Architecture and Hardware Design

. *Four-Layer Functional Architecture*

The proposed embedded architecture is partitioned into four tightly coupled functional layers executed sequentially within a deterministic 1 Hz (1000 ms) real-time loop on the ATmega328P microcontroller:

- **Layer 1 (Multi-Modal Data Acquisition):** Samples 10-bit analog voltages (0–5 V, 0–1023 ADC counts) from the MQ2, MQ5, and MQ7 gas sensors on pins `A0`–`A2`, active-low digital infrared flame status on pin `D4`, digital 1-Wire ambient temperature (°C) and relative humidity (%RH) from the DHT22 on pin `D3`, and user tactile input on pin `D7`.
- **Layer 2 (Dual-Stage EMA Conditioning & Normalization):** Executes startup 50-sample clean-air auto-calibration, applies a recursive low-pass Signal EMA filter ($\alpha_{\text{signal}} = 0.20$) to raw gas channels, computes dimensionless normalized deviations relative to dynamic baselines, and updates the baselines via a conditional slow EMA ($\alpha = 0.01$) during verified safe periods.
- **Layer 3 (Sensor Fusion & Temporal Confirmation):** Maps the 3D normalized vector $\mathbf{X}(t)$ to four distinct hazard signatures (Carbon Monoxide, LPG/Propane, Smoke/Fire, and Methane/NG) and enforces a 2000 ms persistence timer before confirming a candidate class.
- **Layer 4 (Autonomous Mitigation & ML Telemetry):** Drives the buzzer (`D5`), switching transistor fan stage (`D6`), and state-buffered 16×2 I2C LCD (`0x27` on `A4`/`A5`), and transmits a 17-feature CSV record over UART Serial (9600 baud).

The end-to-end signal flow across all four architectural layers is illustrated in Fig. 1.

. *Hardware Pin Mapping and Protection Circuitry*

Table II details the complete hardware interconnections and discrete protection components. Because the DHT22 communicates over an open-drain 1-Wire protocol, an external 10 kΩ pull-up resistor is connected between the `DATA` line (`D3`) and the +5 V rail to guarantee crisp logic-high transitions.

To prevent over-current damage to the ATmega328P GPIO pin (`D6`, rated at 40 mA absolute maximum) when energizing the 5 V brushless DC exhaust fan, a low-side NPN transistor switching circuit is employed. Pin `D6` drives the base of a 2N2222 NPN transistor through a 1 kΩ current-limiting resistor; the emitter is tied to common `GND`, and the collector sinks current from the negative terminal of the fan (whose positive terminal connects to +5 V). Crucially, a 1N4007 rectifier diode is wired in reverse bias across the fan terminals (cathode/stripe to +5 V, anode to the 2N2222 collector) as a flyback snubber to clamp high-voltage inductive kickback transients ($V_L = -L\,di/dt$) when the fan turns off.

. Mathematical Formulation and Signal Conditioning

. *Startup Clean-Air Auto-Calibration*

Upon power-up, after displaying `WARMING UP` to allow thermal stabilization of the SnO_2 heaters, the firmware transitions to the `INITIALIZING` state and records $N = 50$ consecutive clean-air samples spaced by $\Delta t = 100$ ms (5.0 s total window). For each sensor $k \in \{\text{MQ2}, \text{MQ5}, \text{MQ7}\}$, the initial baseline $S_{\text{base},k}(0)$ and initial filtered state $S_{\text{filt},k}(0)$ are initialized to the sample mean:

$$S_{\text{base},k}(0) = \frac{1}{N}\sum_{i=1}^{N} S_{\text{raw},k}(i), \quad S_{\text{filt},k}(0) = S_{\text{base},k}(0), \quad N = 50 \tag{1}$$

Seeding $S_{\text{filt},k}(0) = S_{\text{base},k}(0)$ eliminates cold-start filter convergence lag during the first monitoring cycles.

. *Stage 1: Fast Signal EMA Noise Filtering*

During continuous operation at discrete time steps t ($\Delta T = 1000$ ms), raw 10-bit ADC readings $S_{\text{raw},k}(t)$ are passed through a first-order recursive Exponential Moving Average (EMA) low-pass filter with smoothing factor $\alpha_{\text{signal}} = 0.20$:

$$S_{\text{filt},k}(t) = \alpha_{\text{signal}} S_{\text{raw},k}(t) + \left(1 - \alpha_{\text{signal}}\right) S_{\text{filt},k}(t-1) \tag{2}$$

Table I

COMPARATIVE EVALUATION OF MICROCONTROLLER-BASED GAS AND FIRE DETECTION ARCHITECTURES

Reference	Sensing Array	Signal Conditioning	Drift Tracking	Gas Classes	ML Telemetry & Actuation Capabilities
Baballe *et al.* [1]	Single MQ5	None (Raw ADC)	Static threshold	1 (Generic LPG)	Buzzer & LCD only; no fan or ML data logging
Islam *et al.* [2]	Gas + Flame	Raw threshold	None	Binary Alert	IoT alert & exhaust fan; no EMA or CSV feature stream
Adekitan *et al.* [3]	LPG Sensor	Raw comparator	None	1 (LPG)	Buzzer, fan, solenoid & SMS; no multi-gas classification
Ayaz & Yüksel [4]	MQ Array	Chamber calibration	Manual zeroing	CH_4, H_2	Experimental characterization; no autonomous fan/ML stream
Apeh *et al.* [5]	Kitchen MQ	Fixed voltage ref	None	1 (LPG)	Shutoff valve & ventilator; no temporal debouncing
Khan [6]	Single MQ	Raw threshold	None	Binary Alert	Household alarm & control; lacks multi-sensor fusion
Proposed System	**MQ2, MQ5, MQ7, Flame, DHT22**	**Fast Signal EMA ($\alpha_{sig} = 0.20$) + Norm**	**Conditional EMA ($\alpha = 0.01$, $\lvert S_{norm} \rvert < 0.05$)**	**4 Classes + 2000 ms Debounce**	**Flyback-protected fan, anti-flicker I2C LCD, buzzer, & 1 Hz 17-col ML CSV stream**

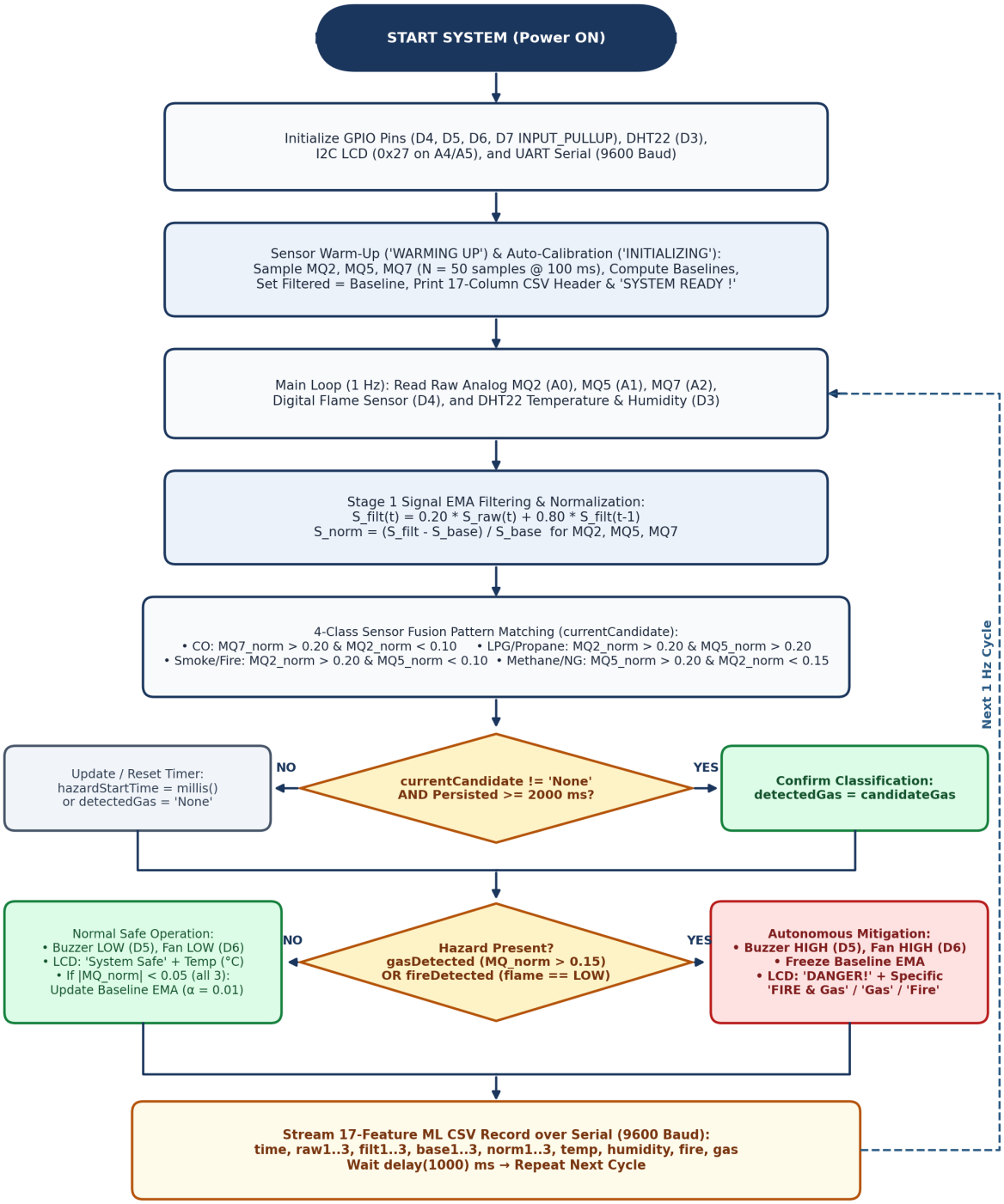


Fig. 1. System architecture and block diagram of the multi-sensor adaptive electronic nose, autonomous mitigation stages, and 17-feature ML telemetry pipeline.

Table II
HARDWARE PIN MAPPING AND PROTECTION SPECIFICATIONS

Module	Module Pin	MCU Pin	Electrical Configuration
MQ2 Sensor	AO, VCC, GND	`A0`	10-bit ADC; LPG, C_3H_8, H_2, Smoke
MQ5 Sensor	AO, VCC, GND	`A1`	10-bit ADC; Methane, NG, LPG
MQ7 Sensor	AO, VCC, GND	`A2`	10-bit ADC; Carbon Monoxide (CO)
DHT22	DATA, VCC, GND	`D3`	10 kΩ pull-up (DATA ↔ 5V)
Flame Sensor	DO, VCC, GND	`D4`	Active-LOW digital IR output
Buzzer	IN, VCC, GND	`D5`	Active-HIGH piezoelectric alarm
Fan Driver	2N2222 Base	`D6`	Via 1 kΩ resistor + 1N4007 diode
Push Button	Terminal 1, 2	`D7`	`INPUT_PULLUP` (other side GND)
I2C LCD	SDA, SCL	`A4, A5`	16×2 LCD at I2C address `0x27`

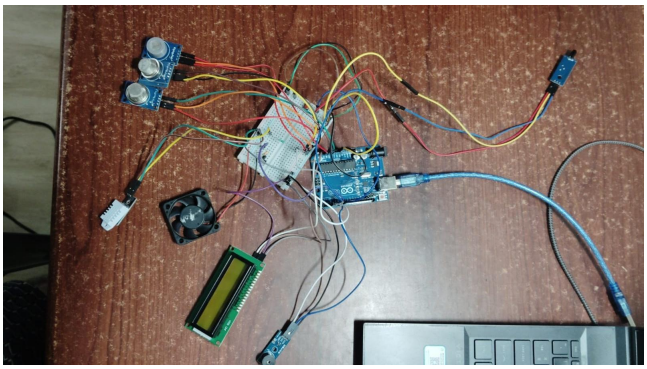


Fig. 2. Complete experimental hardware prototype showing the Arduino Uno R3, MQ2/MQ5/MQ7 sensor array, DHT22 with 10 kΩ pull-up, IR flame detector, 2N2222 fan driver with 1N4007 flyback diode, active buzzer, and 16×2 I2C LCD.

With $\alpha_{\text{signal}} = 0.20$, random single-sample ADC spikes are attenuated by 80% on arrival, while genuine gas concentration steps reach $> 89\%$ of their steady-state amplitude within 10 samples, preserving rapid hazard responsiveness.

. *Relative Normalization and Stage 2 Conditional Baseline EMA*

To eliminate inter-sensor offset differences and convert filtered ADC counts into dimensionless fractional resistance changes, each sensor's filtered reading is normalized relative to its current baseline:

$$S_{\text{norm},k}(t) = \frac{S_{\text{filt},k}(t) - S_{\text{base},k}(t-1)}{S_{\text{base},k}(t-1)} \tag{3}$$

To counteract slow baseline drift caused by ambient temperature shifts, humidity variations, and sensor aging without allowing a slow gas leak to be absorbed into the baseline, the firmware implements a *conditional slow EMA baseline tracker* with adaptation rate $\alpha = 0.01$. Let $H_{\text{gas}}(t)$ and $H_{\text{fire}}(t)$ denote the boolean gas and fire hazard flags:

$$H_{\text{gas}}(t) = \bigvee_{k \in \{\text{MQ2,MQ5,MQ7}\}} \Big(S_{\text{norm},k}(t) > 0.15 \Big) \tag{4}$$

$$H_{\text{fire}}(t) = \Big(D_{\text{flame}}(t) == \text{LOW} \Big) \tag{5}$$

The baseline $S_{\text{base},k}(t)$ updates if and only if no hazard is active and all three sensors reside strictly within a $\pm 5\%$ quiescent deadband:

$$S_{\text{base},k}(t) = \begin{cases} (1-\alpha) S_{\text{base},k}(t-1) + \alpha S_{\text{raw},k}(t), & \text{if } \neg H_{\text{gas}}(t) \wedge \neg H_{\text{fire}}(t) \\ & \text{and } \max_j |S_{\text{norm},j}(t)| < 0.05 \\ S_{\text{base},k}(t-1), & \text{otherwise (Frozen)} \end{cases} \tag{6}$$

Key Signal Conditioning Insight: By gating the $\alpha = 0.01$ baseline update with the strict condition $\max_j |S_{\text{norm},j}(t)| < 0.05$, the electronic nose tracks diurnal thermal/humidity drift during clean-air operation but immediately locks the reference baseline whenever any sensor deviates by $\geq 5\%$.

. TEMPORAL SENSOR FUSION AND ML TELEMETRY PIPELINE

. *Four-Class Deterministic Pattern Fusion*

Because MQ2, MQ5, and MQ7 exhibit distinct relative sensitivities across different gases, the firmware constructs the feature vector $\mathbf{X}(t) = [\text{MQ2}_{\text{norm}}(t), \text{MQ5}_{\text{norm}}(t), \text{MQ7}_{\text{norm}}(t)]^T$ and evaluates a hierarchical 4-class rule set to assign an instantaneous candidate label $C_{\text{cand}}(t)$:

$$C_{\text{cand}}(t) = \begin{cases} \texttt{Carbon Monoxide}, & \text{if MQ7}_{\text{norm}} > 0.20 \wedge \text{MQ2}_{\text{norm}} < 0.10 \\ \texttt{LPG/Propane}, & \text{if MQ2}_{\text{norm}} > 0.20 \wedge \text{MQ5}_{\text{norm}} > 0.20 \\ \texttt{Smoke/Fire}, & \text{if MQ2}_{\text{norm}} > 0.20 \wedge \text{MQ5}_{\text{norm}} < 0.10 \\ \texttt{Methane/NG}, & \text{if MQ5}_{\text{norm}} > 0.20 \wedge \text{MQ2}_{\text{norm}} < 0.15 \\ \texttt{None}, & \text{otherwise} \end{cases} \tag{7}$$

. *Temporal Confirmation (Debounce) Filter*

To prevent brief airflow disturbances or sensor transients from causing erratic class switching, $C_{\text{cand}}(t)$ must remain invariant for at least $T_{\text{confirm}} = 2000$ ms (tracked via the hardware timer `millis() - hazardStartTime >= confirmationTime`) before being promoted to the confirmed output class $C_{\text{det}}(t)$ (`detectedGas`):

$$C_{\text{det}}(t) = \begin{cases} C_{\text{cand}}(t), & \text{if } C_{\text{cand}}(t) \neq \texttt{None} \wedge (t - t_{\text{start}}) \geq 2 \text{ s} \\ \texttt{None}, & \text{if } C_{\text{cand}}(t) == \texttt{None} \end{cases} \tag{8}$$

. *Anti-Flicker LCD Buffering and Priority Actuation*

When $H_{\text{gas}}(t) \vee H_{\text{fire}}(t)$ evaluates to true, the microcontroller drives `buzzerPin` (`D5`) and `fanPin` (`D6`) `HIGH` and selects one of three priority-ordered warning strings on the 16×2 I2C LCD: (1) `"DANGER! / FIRE & " + detectedGas` if both gas and optical flame are present, (2) `"DANGER! / Gas: " + detectedGas` for gas-only events, or (3) `"DANGER! / Fire Detected!"` for flame-only events. To eliminate the visual flicker caused by calling `lcd.clear()` every 1000 ms, the helper function `updateLCD(line1, line2)` caches the previous frame (`lastLine1`, `lastLine2`) and writes to the I2C bus only upon a state transition.

. *Production 17-Feature ML CSV Telemetry Stream*

At the conclusion of every 1000 ms cycle, the firmware serializes a 17-column comma-delimited record over UART (9600 baud), as specified in Table III. The complete real-time execution flow is shown in Fig. 3, and the open-source firmware implementation is available on GitHub.

Table III
PRODUCTION 17-FEATURE MACHINE LEARNING CSV TELEMETRY SCHEMA

Col	CSV Header	Format	Feature Description & ML Role
1	`time`	`ulong` (ms)	Monotonic boot timestamp (`millis()`)
2–4	`raw1..raw3`	`int` (0–1023)	Unfiltered ADC counts (MQ2, MQ5, MQ7)
5–7	`filt1..filt3`	`float` (1 dec)	Fast Signal EMA values ($\alpha_{\text{signal}} = 0.20$)
8–10	`base1..base3`	`float` (1 dec)	Adaptive clean-air baselines ($\alpha = 0.01$)
11–13	`norm1..norm3`	`float` (4 dec)	Normalized deviations; primary ML features
14	`temp`	`float` (°C)	Ambient temperature from DHT22 (D3)
15	`humidity`	`float` (%RH)	Relative humidity from DHT22 (D3)
16	`fire`	`int` (0/1)	Binary IR flame flag (1 if `flame==LOW`)
17	`gas`	`String`	Confirmed class label (`detectedGas`)

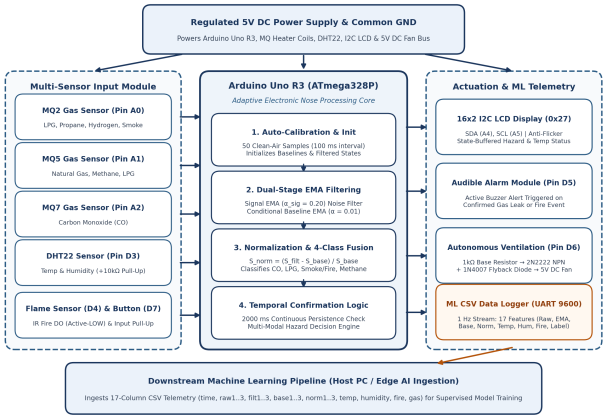


Fig. 3. Algorithmic flowchart of the embedded firmware, illustrating the 50-sample startup auto-calibration, fast Signal EMA filter ($\alpha_{\text{signal}} = 0.20$), 4-class pattern fusion, 2000 ms temporal confirmation branch, conditional Baseline EMA drift tracker ($\alpha = 0.01$), and 1 Hz CSV telemetry loop.

. EXPERIMENTAL EVALUATION AND DISCUSSION

. *Experimental Testbed and Gas Exposure Protocol*

The hardware prototype was assembled on a solderless breadboard interfaced with the Arduino Uno R3 and evaluated across four controlled physical test cases: (1) ambient clean air (25.6 °C, 58.2% RH), (2) butane/LPG exposure from an unlit gas source, (3) cellulose combustion smoke and direct flame, and (4) carbon monoxide-rich plumes generated by smoldering paper/incense. Both the production 17-column CSV stream (Table IV) and bench validation metrics (Table V) were recorded via the Arduino IDE Serial Monitor at 9600 baud.

. *Signal Conditioning and Classification Performance*

Under quiescent clean-air conditions (rows 6012 ms and 7014 ms in Table IV), normalized deviations across all three MQ channels remained within ± 0.0014 (and within -0.02 to -0.04 during extended bench runs), well inside the $|S_{\text{norm}}| < 0.05$ deadband. Consequently, the conditional slow EMA filter ($\alpha = 0.01$) continuously tracked slow ambient drift while maintaining `detectedGas = None` and displaying `System Safe / Temp: 25.60C` on the LCD.

The transient noise rejection, baseline adaptation, and 2000 ms confirmation behavior are illustrated in Fig. 4.

During Carbon Monoxide exposure (smoldering incense/paper), MQ7_{norm} rose above the 0.20 classification threshold (reaching 0.22–0.42) while MQ2_{norm} remained below 0.10 (-0.03 to 0.09). As illustrated between $t = 10019$ ms and $t = 12023$ ms in Table IV, the temporal confirmation filter held the confirmed label at `None` during the initial 2000 ms persistence window before locking in `Carbon Monoxide`, activating the buzzer and 2N2222 exhaust fan driver, and freezing the clean-air baselines at $[218.0, 194.0, 165.0]$. Similarly, during LPG/Propane exposure, simultaneous elevation of $\text{MQ2}_{\text{norm}} \in [0.22, 0.43]$ and $\text{MQ5}_{\text{norm}} \in [0.32, 0.54]$ with low $\text{MQ7}_{\text{norm}} \approx 0.02$ accurately triggered the `LPG/Propane` state.

. *Current Limitations and Future Research Directions*

While the dual-stage EMA architecture and 2000 ms temporal debounce filter resolve baseline drift and transient noise, three engineering extensions remain for future work:

1. *Supervised Edge-ML Deployment:* Training Random Forest, Support Vector Machine (SVM), or lightweight 1D-CNN classifiers on the logged 17-feature CSV dataset to decouple simultaneous multi-gas mixtures and deploying inference on a Raspberry Pi or ESP32 edge gateway.
2. *On-Board Thermodynamic Compensation:* Utilizing the logged DHT22 `temp` and `humidity` channels to apply non-linear R_s/R_0 temperature-humidity correction surfaces directly on the microcontroller prior to threshold evaluation.
3. *Hardware Interrupt Calibration & PWM Ventilation:* Binding the push button on pin D7 (`INPUT_PULLUP`) to an interrupt service routine for one-touch baseline reset and manual ML ground-truth labeling, and upgrading pin D6 from binary switching to Proportional-Integral-Derivative (PID) PWM fan speed control.

. CONCLUSION

This paper presented an adaptive multi-sensor fire and gas hazard mitigation system with native Machine Learning telemetry logging on the Arduino Uno R3. By integrating 50-sample startup auto-calibration, fast Signal EMA noise filtering ($\alpha_{\text{signal}} = 0.20$), conditional slow Baseline EMA drift compensation ($\alpha = 0.01$), 4-class pattern fusion, and a 2000 ms temporal persistence filter, the system achieves robust local hazard mitigation via a flyback-protected 2N2222 exhaust fan driver, audible buzzer, and anti-flicker I2C LCD while streaming a 17-feature CSV dataset at 1 Hz for supervised ML modeling.

REFERENCES

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Table IV

REPRESENTATIVE 17-COLUMN UART SERIAL CSV TELEMETRY STREAM CAPTURED ACROSS EXPERIMENTAL REGIMES (1 HZ, 9600 BAUD)

time	raw1	raw2	raw3	filt1	filt2	filt3	base1	base2	base3	norm1	norm2	norm3	temp	hum	fire	gas (Confirmed Label & State)
6012	218	194	165	218.2	194.1	165.0	218.0	194.0	165.0	0.0009	0.0005	0.0000	25.6	58.2	0	None (Quiescent Clean Air)
7014	217	195	164	218.0	194.3	164.8	218.0	194.0	165.0	−0.0002	0.0014	−0.0012	25.6	58.2	0	None (Baseline EMA Active)
10019	230	214	276	224.2	202.5	206.4	218.0	194.0	165.0	0.0284	0.0438	0.2509	25.7	58.0	0	None (CO Candidate < 2 s)
12023	232	216	288	226.8	207.2	235.1	218.0	194.0	165.0	0.0404	0.0680	0.4248	25.7	57.9	0	Carbon Monoxide (≥ 2 s)
14027	345	332	168	263.8	246.2	211.6	218.0	194.0	165.0	0.2101	0.2691	0.2824	25.8	57.5	0	None (LPG Candidate < 2 s)
16031	364	354	165	298.9	284.1	195.0	218.0	194.0	165.0	0.3711	0.4644	0.1818	25.9	57.2	0	LPG/Propane (≥ 2 s)
17033	368	359	165	312.7	299.1	189.0	218.0	194.0	165.0	0.4344	0.5418	0.1455	26.4	56.8	1	LPG/Propane (FIRE & LPG)

Table V

MULTI-SENSOR NORMALIZED RESPONSE MATRIX, TEMPORAL CONFIRMATION STATE, AND AUTONOMOUS MITIGATION OUTPUT ACROSS OPERATING REGIMES

Operating Regime	$MQ2_{norm}$	$MQ5_{norm}$	$MQ7_{norm}$	Flame (D4)	CSV Class (gas)	Actuator State, Baseline EMA Status, & I2C LCD Message
Clean Air (Safe)	< 0.05	< 0.05	< 0.05	HIGH (0)	None	Buzzer LOW, Fan LOW, Baseline EMA Adapting; LCD: System Safe / Temp: 25.60C
Carbon Monoxide	< 0.10	0.10–0.17	> 0.20	HIGH (0)	Carbon Monoxide	Buzzer HIGH, Fan HIGH, Baseline Frozen; LCD: DANGER! / Gas: Carbon Mono
LPG / Propane Leak	> 0.20	> 0.20	< 0.05	HIGH (0)	LPG/Propane	Buzzer HIGH, Fan HIGH, Baseline Frozen; LCD: DANGER! / Gas: LPG / Propa
Smoke / Incipient Fire	> 0.20	< 0.10	0.05–0.18	HIGH (0)	Smoke/Fire	Buzzer HIGH, Fan HIGH, Baseline Frozen; LCD: DANGER! / Gas: Smoke / Fir
Methane / Natural Gas	< 0.15	> 0.20	< 0.05	HIGH (0)	Methane/NG	Buzzer HIGH, Fan HIGH, Baseline Frozen; LCD: DANGER! / Gas: Methane/NG
Direct Flame (+ Gas)	Variable	Variable	Variable	LOW (1)	<detectedGas>	Buzzer HIGH, Fan HIGH, Baseline Frozen; LCD: DANGER! / FIRE & <Gas> or Fire Detected!

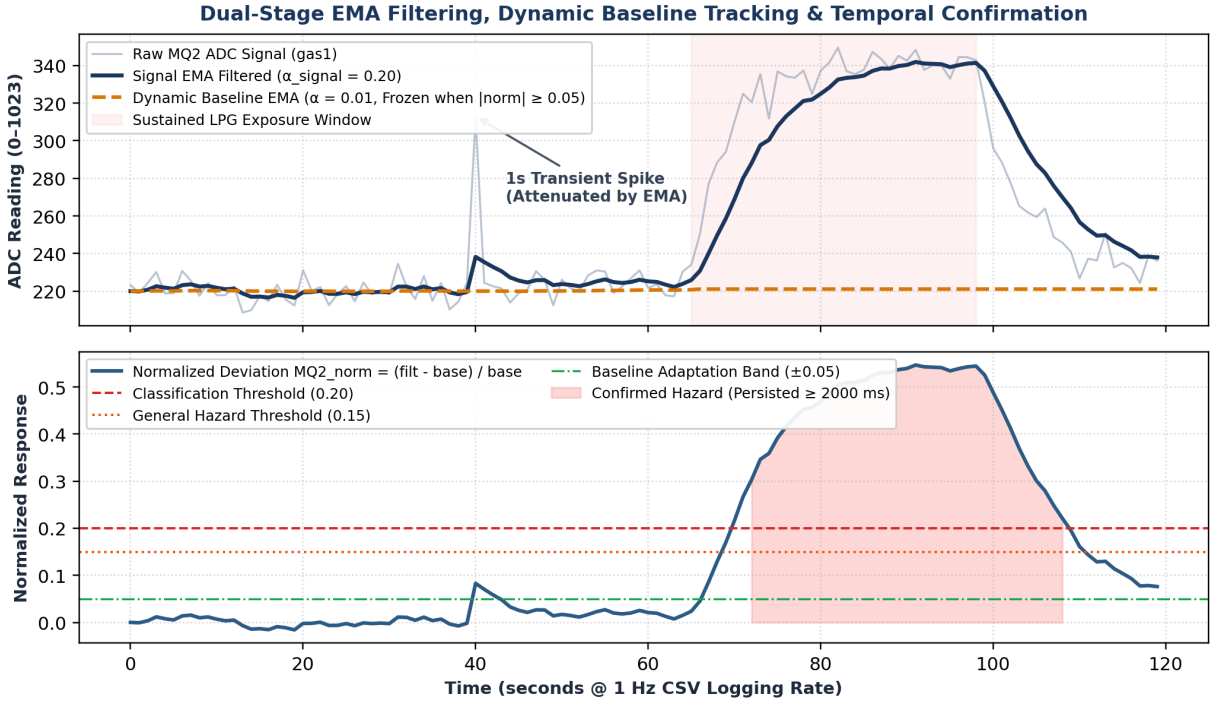


Fig. 4. Dual-stage EMA signal conditioning and temporal confirmation response: (top) raw MQ2 ADC signal versus fast Signal EMA ($\alpha_{\text{signal}} = 0.20$) and conditional dynamic Baseline EMA ($\alpha = 0.01$, frozen when $|S_{\text{norm}}| \geq 0.05$); (bottom) normalized deviation showing attenuation of a 1 s transient spike and confirmed detection of a sustained LPG leak after 2000 ms.

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